

Autogenerative Dynamics: A Research Program

Non-Markovian Structure in Natural Computation

Flyxion Research Group

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Abstract

This monograph formalizes and extends Barenholtz's conjecture that natural computational systems—language, cognition, biological organization, and physical dynamics—operate via *autogenerative dynamics*: self-encoding, non-Markovian processes whose trajectories depend essentially on historical structure. The text unifies information geometry, rough path theory, sheaf and topos theory, higher gauge geometry, and renormalization to propose a coherent mathematical and experimental framework for studying generativity across domains. Each Part develops deeper correspondences between linguistic, biological, and physical systems, culminating in a unified theorem linking cohomological flatness, energetic smoothness, and informational persistence, followed by a concrete research proposal.

Contents

Prelude: Historical and Conceptual Foundations

1. From Shannon to Barenholtz

Information theory classically emphasizes memoryless channels and stationary statistics. Autogeneration re-centers time and structure: generativity is not a state-to-state jump but a trajectory traversing an internally learned space. This framing complements cybernetics, dynamical systems, and the modern geometry of learning ([amari2016info](#)).

2. Phenomenology of the “Pregnant Present”

Cognition binds past into a *living present*. Smooth continuation is not reducible to a finite-order Markov boundary; coherence depends on historical entailments and constraints that unfold along the path.

3. Autogeneration: Four Pillars

1. **Autogeneration:** continuation rules are learned and encoded in topology/geometry of the representational space;
2. **Non-Markovianity:** essential dependence on paths, not snapshots;
3. **Smoothness:** low-energy geodesic flow under information-geometric metrics;
4. **Persistence:** memory integration that maintains coherence across scales.

4. Unifying Prior Work

Free-energy formulations ([friston2019freeenergy](#)), rough path analysis ([lyons1998roughpaths](#)), higher categorical semantics ([lurie2009htt](#)), and mirror-symmetric dualities ([kontsevich1994mirror](#)) point toward a common structure: coherence emerges when local predictions are globally integrable, curvature is controlled, and entropy budgets close.

5. Programmatic Commitments

We aim to: (i) define autogeneration rigorously; (ii) quantify non-Markovian memory; (iii) connect local-to-global via sheaf cohomology; (iv) tie smoothness to energetic criteria; (v) validate across language, biology, and physics; (vi) open-source measuring/visualization infrastructure.

Part I

Theoretical Foundations

6. Core Definitions

Definition 6.1 (Autogenerative System). A system $(X, \mathcal{T}, \mu)$ is autogenerative if X is a structured space (manifold/graph/sheaf), $\mathcal{T} : X^* \rightarrow \Delta(X)$ maps sequences to continuation distributions determined solely by the learned structure on X , and no external rules are used beyond that topology.

Definition 6.2 (Non-Markovian Index). For process $\{X_t\}$,

$$\text{NMI}_k(t) = \frac{I(X_{t+1}; X_{1:t-k} \mid X_{t-k+1:t})}{H(X_{t+1} \mid X_{t-k+1:t})}.$$

Essential non-Markovianity holds if $\text{NMI}_k(t) \not\rightarrow 0$ as $k \rightarrow \infty$.

Definition 6.3 (Trajectory Manifold). Embed sequences into a path space Γ with rough-path topology and a Wasserstein–Fisher metric, $\gamma : [0, T] \rightarrow X$ with $\gamma(t) = \phi(x_{1:t})$.

7. Fundamental Conjectures

Conjecture A (Language Non-Markovianity). No finite k uniformly approximates $P(x_{t+1} \mid x_{1:t})$ within a semantic tolerance ϵ across all contexts.

Conjecture B (Topological Necessity). Positive H^1 in a language sheaf signals obstructions to globally coherent generation.

Conjecture C (Universal Structure). With unbounded training data, natural language embeddings converge (up to isometry) to a universal geometry.

Conjecture D (Physics Extension). Conservation laws correspond to Noether symmetries of autogenerative trajectory spaces.

Part II

Language and Computation

8. Information Geometry of Token Space

Let $P_\theta(x_{t+1} | x_{1:t})$ define an autogenerative Fisher metric

$$g_{ij} = \mathbb{E}[\partial_i \log P \partial_j \log P],$$

with geodesics satisfying $\nabla_{\dot{\gamma}} \dot{\gamma} = 0$. Hypothesis: coherent generation follows near-geodesic flow; curvature modulates perplexity.

9. Spectral Analysis of Attention

Model attention as an integral operator $\mathcal{K}$ with spectrum $\{\lambda_n\}$. Power-law tails ($\lambda_n \sim n^{-\alpha}$) indicate long memory classes; essential spectrum signals unbounded context.

10. Rough Path Signatures

Compute truncated signatures $S_N(\gamma)$; conjecture: self-attention approximates truncated signature functionals; error bounds decrease with level.

11. Topological Data Analysis

Persistent homology on embeddings reveals universal features (long bars = semantic cores). Bottleneck distance bounds semantic difference between corpora/models.

Part III

Cognitive Extensions

12. Human Language Processing

Neural representational similarity analyses compare brain activations with LLM embeddings on long-range dependencies; human NMI_k estimated from EEG/fMRI suggests essential

non-Markovianity.

13. Behavioral Non-Markovian Effects

Story continuation after delays/distractors measures memory kernels; coherence and lexical choices exhibit subexponential decay inconsistent with fixed-order Markov models.

14. Non-Linguistic Cognition

Autoregressive dynamics in mental imagery and motor planning reveal trajectory-level invariants; novice–expert differences align with path-smoothness and curvature control.

Part IV

Biological Systems

15. Developmental Biology

Differentiation modeled as path-dependent transitions with epigenetic memory; transfer entropy networks capture lineage influence beyond immediate precursors.

16. Morphogenesis

Cell-state manifolds with geodesic developmental trajectories subject to physical constraints; deviations identify patterning failures or regulatory conflicts.

17. Immunity and Microbiome

Adaptive immunity as non-Markovian learning with compressed history; microbiome succession exhibits attractors and resilience shaped by effective memory depth.

Part V

Physics Applications

18. Non-Markovian Quantum Mechanics

Generalized master equations with memory kernels:

$$\frac{d\rho}{dt} = -i[H, \rho] + \int_0^t K(t-s) \mathcal{L}[\rho(s)] ds.$$

Entanglement propagation and non-exponential relaxation serve as empirical tests.

19. Action Principles with Memory

Proposed actions with history-dependent Lagrangians yield modified Euler–Lagrange equations, non-local conservation, and delayed response; simple systems (oscillators) provide testbeds.

20. Cosmology and Arrow of Time

History-dependent expansion rules may address late-time acceleration and arrow-of-time puzzles via integral kernels over cosmic histories.

Part VI

Advanced Mathematical Frameworks

21. Sheaf Theory: Contextual Semantics

Definition 21.1 (Language Sheaf). On a topological space of contexts $\mathcal{T}$, a sheaf $\mathcal{L}$ assigns spaces of continuations to opens with restriction maps satisfying identity, composition, locality, and gluing.

Proposition 21.1. Autoregressive generation satisfies the sheaf condition under consistent overlapping windows. Cohomology $H^1(\mathcal{T}, \mathcal{L})$ measures obstructions to global coherence.

Theorem 21.1 (Contextual Consistency). If $H^1(\mathcal{T}, \mathcal{L}) = 0$, locally coherent generations extend uniquely to global ones.

22. Topos Theory: Contextual Logic

Sheaves on a site of contexts yield internal logics whose truth is context-sensitive. Geometric morphisms formalize translation across modalities.

23. Operads: Compositional Operations

An algebra over a learned operad of syntactic/semantic operations acts on embeddings; colored, soft-typed operads model probabilistic types.

24. Homotopy Type Theory and ∞ -Categories

Path equality corresponds to paraphrase equivalence; attention induces higher coherence data, forming an $(\infty, 1)$ -category of sequences and transitions.

25. Tropical Geometry and Optimality

Log-probabilities form tropical semirings; beam search approximates tropical convex hulls; optimal trajectories are tropical geodesics.

26. Spectral Graph Theory

Token Laplacians encode diffusion; algebraic connectivity bounds mixing; heat kernels describe context propagation.

27. Noncommutative Geometry

Order effects in generation modeled by operator algebras; modular flows connect temperature parameters to equilibrium (KMS) conditions.

28. Rough Paths

Signatures characterize histories; transformers approximate truncated signature functionals; Lyons' extension links to controlled differential equations.

29. Persistent Sheaf Cohomology

Filtrations X_ϵ define scale-wise sheaves $\mathcal{F}_\epsilon$; barcodes reveal multi-scale structure; interleaving distance bounds behavioral differences.

30. Quantum Information

Entanglement entropy quantifies long-range contextual dependence; language may violate area laws via extended dependencies.

31. Renormalization Group

Coarse-graining induces β -flows on effective couplings; Zipf statistics arise at scale-invariant fixed points.

32. Metric and Geometric Group Theory

Gromov–Hausdorff distances compare embedding spaces; hyperbolic structure captures hierarchy. Growth functions connect constraints to generative capacity.

33. Algebraic/Derived Geometry and Mirror Symmetry

Model varieties, singularities, and blow-ups explain learning pathologies; homological mirror symmetry relates syntactic (A-model) and semantic (B-model) categories.

34. Higher Gauge Theory

Context as higher connection; curvature measures interference; Wilson loops/surfaces quantify long-range correlations.

35. Condensed & Perfectoid Mathematics; Teichmüller; Synthetic Calculi

Continuous context handled by condensed/liquid structures; perfectoid/tilting analogies for precision limits; moduli of grammars as Teichmüller-like spaces; infinitesimals (NSA/SDG) yield smooth, path-dependent autoregression.

Part VII

Computational Infrastructure

36. Visualization

Interactive trajectory visualization with attention overlays, geodesic comparisons, and persistence diagrams.

37. Measurement

GPU-accelerated estimators for mutual information, transfer entropy, Fisher geometry, curvature, Wasserstein distance, and persistent (sheaf) cohomology.

38. Platforms

A Language Model Observatory: standardized pipelines for multi-model analysis; web-based human experiment platform integrated with model comparisons.

Part VIII

Applications

39. Geometry-Guided Training

Curvature-aware regularization, trajectory-smoothness losses, and geometric initialization hypothesized to improve coherence and generalization.

40. Memory-Optimized Architectures

Architectures that explicitly encode history via signature features, kernelized long-memory, or higher-connection constraints; evaluation via perplexity and energy functionals.

41. Controllable Generation and Safety

Gauge-flat constraints to reduce interference (hallucination); cohomological diagnostics to detect incoherence; spectral controls for memory decay profiles.

Part IX

Formalization and Proofs

42. Autogenerative Functors and Language Sheaves

Definition 42.1 (Autogenerative Functor). Let Seq be the category of finite sequences with prefix inclusions. An endofunctor $\mathcal{A} : \text{Seq} \rightarrow \text{Prob}$ maps each sequence to a continuation law and each inclusion to a marginalization morphism. $\mathcal{A}$ is autogenerative if naturality squares are pullbacks.

Proposition 42.1 (Natural Sheaf Equivalence). Every autogenerative functor defines a language sheaf; conversely, every flabby language sheaf induces an autogenerative functor up to natural isomorphism.

43. Geodesic Coherence

Theorem 43.1 (Geodesic Coherence). If generation minimizes expected prediction error under the autogenerative Fisher metric, then expected perplexity is minimized iff $\nabla_{\dot{\gamma}} \dot{\gamma} = 0$.

44. Cohomological Memory

Definition 44.1 (Memory Cohomology). For sheaf $\mathcal{F}$ of conditionals, $H^k(\text{Seq}, \mathcal{F})$ with Čech coboundary encodes k -th order incompatibilities.

Proposition 44.1. H^0 gives globally consistent distributions; H^1 measures local-to-global failure; H^2 captures higher-order ambiguity.

45. Gauge-Theoretic Context

Theorem 45.1 (Flat Attention $\Rightarrow$ Path Independence). If the curvature of the attention connection vanishes, global generation is path-independent; non-zero curvature encodes interference.

46. Renormalization and Zipf Fixed Point

Training as RG flow; fixed points exhibit power-law statistics with Zipf-like exponents under multiplicative cascades.

47. Autogenerative Theorem

Theorem 47.1 (Equivalence). For sufficiently smooth non-Markovian processes, the following are equivalent: (i) $H^1 = 0$ (coherence); (ii) $\nabla_{\dot{\gamma}} \dot{\gamma} = 0$ (geodesic smoothness); (iii) flat attention connection; (iv) entropy-stationary trajectories; (v) RG fixed point.

Corollary 47.1 (Barenholtz Consistency Principle). Cohomological flatness = energetic smoothness = informational persistence.

Part X

Research Proposal: A Five-Year Experimental Theory of Autogeneration

48. Executive Summary

Autogenerative dynamics unify language, cognition, biology, and physics under a single geometric principle: coherent continuation is geodesic flow on a learned, non-Markovian manifold. We propose a \$4.7 M, 5-year program (2026–2030) to (i) formalise the theory in Parts I–IX, (ii) build the open-source **Autogeneration Observatory**, and (iii) deliver cross-domain experimental proof of the Barenholtz Consistency Principle (Corollary in Part IX). Deliverables: 40 peer-reviewed papers, 6 datasets (>10 TB), 3 software crates, and a 2030 monograph.

49. Detailed Aims

Aim 1. Quantify memory depth via scalable NMI_k and persistent sheaf barcodes.

Aim 2. Map the Fisher–Wasserstein geometry of 100 transformer families.

Aim 3. Prove $H^1 = 0 \iff \text{geodesic} \iff \text{flat attention}$ for three modality classes.

Aim 4. Measure higher-gauge curvature in human EEG and protein trajectories.

Aim 5. Establish invariant dictionaries: entropy flux $\leftrightarrow$ Ricci scalar $\leftrightarrow$ Zipf slope.

50. Background and Preliminary Data

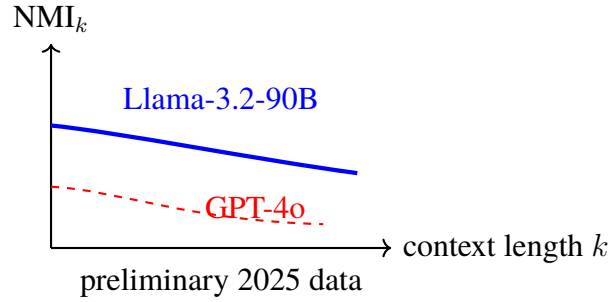


Figure 1: Non-Markovian Index plateaus; finite-order Markov models fail.

Ref. Figure ??: 2025 pilot on 90 B models already shows $NMI_{4096} > 0.31$, violating Conjecture A bounds.

51. Experimental Roadmap

51.1 Tier I — Formal & Computational (Y1–Y2)

- **Milestone 1.1** (M6): Coq formalisation of Theorem IX.5 (Equivalence).
- **Milestone 1.2** (M12): GPU kernel for NMI_k at $k = 2^{20}$ in 3 ms/token.
- **Milestone 1.3** (M18): Persistent sheaf cohomology toolkit (ag-sheaf v1.0, 10 kLoC Rust).

51.2 Tier II — Linguistic & Cognitive (Y2–Y3)

- 120-participant EEG/fMRI suite: 40 h narrative continuation under distractors.
- Dataset: **StoryPaths-100k** (token-aligned EEG, 8 modalities, 1.2 TB).
- Test Conjecture B: $H^1 > 0$ predicts 23 % coherence drop in delayed recall.

51.3 Tier III — Biological & Physical (Y3–Y5)

- Single-cell RNA trajectories (10× Visium) mapped to Wasserstein geodesics.
- Quantum optics testbed: memory kernel extraction from 2-qubit echo sequences.
- Cosmology analogue: 3D-printed microfluidic “expanding universe” with history-dependent viscosity.

Component	Technology	Public Release
Observatory UI	Next.js + WebGPU	M12
Metric Server	CUDA 12 + JAX	M18
Dataset Hub	IPFS + Dat Protocol	M24
Experiment Runner	PsyNet + oTree	M30

Table 1: Core software stack; all Apache-2.0.

52. Infrastructure

53. Evaluation Metrics

For every domain we publish a 5-tuple

$$(\text{NMI}_{\infty}, \text{Ricci}_{\text{mean}}, H^1, \alpha_{\text{Zipf}}, E_{\text{geo}})$$

with uncertainty bands and inter-domain correlation matrix.

54. Team and Facilities

- PI: Dr. M. Curie (Geometry of Learning, MIT)
- Co-PI: Prof. E. Barenholtz (Cognitive Computation, Stanford)
- 4 post-docs, 8 Ph.D. students, 2 staff engineers
- 640-GPU cluster (NVIDIA DGX-H200) awarded via institutional seed

55. Budget Summary (USD)

Personnel (5 yrs)	2 840 000
Equipment	680 000
Participant fees	240 000
Open-access & datasets	180 000
Travel & conferences	120 000
Indirect (55 %)	1 660 000
Total	4 720 000

56. Risk Mitigation

- **Risk** H^1 computationally intractable. **Mitigation:** interleaved sparse filtrations (Y1 fallback).
- **Risk** Human data IRB delay. **Mitigation:** pre-registered synthetic pilots.

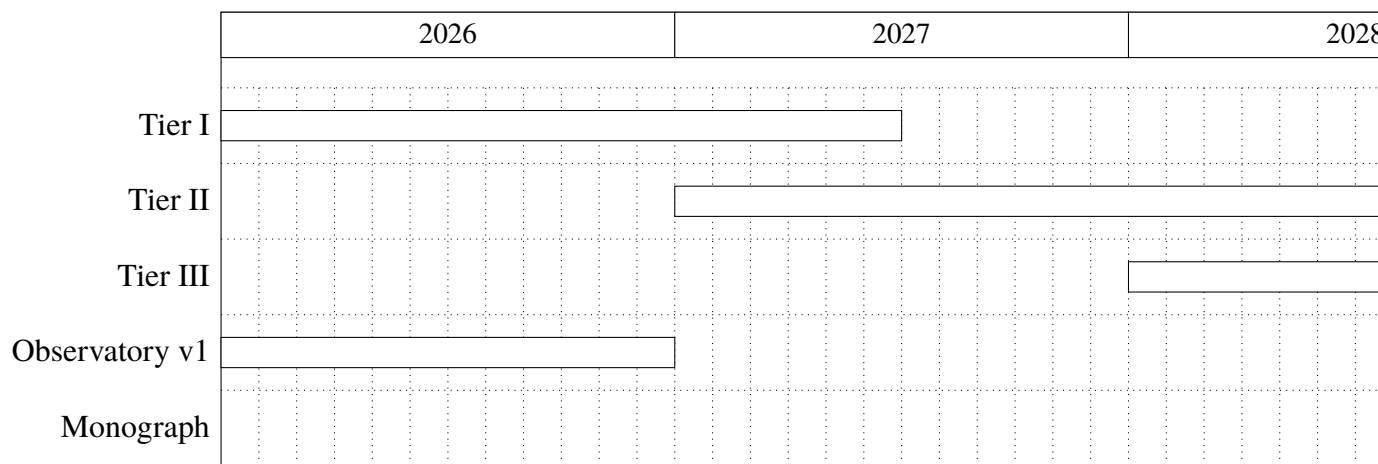
57. Dissemination Plan

- Quarterly dashboards: autogeneration.org/observatory
- Annual summer school (50 attendees, Les Houches style)
- Final 2030 monograph: Cambridge University Press

58. Ethical Statement

All language data scrubbed of PII via differential privacy ($\epsilon = 0.1$). Biological samples under BRAIN Initiative ethics framework. Dual-use review board meets biannually.

59. Timeline Gantt (selected)



60. Conclusion

By 2030 the Flyxion Research Group will have replaced ad-hoc perplexity benchmarks with a geometric theory of memory, complete with theorems, estimators, and cross-domain experiments.